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Learning the Task or Learning the Verifier? Rethinking Criteria Exposure in Embodied Self-Improvement

Task or Verifier? Rethinking Criteria Exposure in Embodied Self-Improvement

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TL;DR

TL;DR: CE's extra gains mainly reflect learning the verifier

Letting the learner read the scoring code (Criteria Exposure, CE) lifts RoboDojo scores, but nearly all extra gain comes from **hidden rules not in the instructions**: it vanishes under instruction-faithful criteria and does not transfer to new tasks.

w/ CE library (reads criteria)

51.6% → 45.0%

16 criteria-limited tasks: re-scoring the same episodes with instruction-faithful C_faith drops 6.6 points

w/o CE library (success only)

27.8% → 47.2%

Re-scoring the same episodes with C_faith adds 19.4 points; the original 23.8-point gap reverses to −2.2

Library learned from C_faith

65.0% → 13.5%

10 tasks: re-scored with the original C_orig, it drops 51.5 points; scores track only the criteria it read

Background

Root cause: benchmark criteria often exceed the instructions

Instruction given to the agent (press_by_number)

"Press the two red buttons the required number of times according to the number cards, then press the blue button to confirm."

Press the two red buttons as many times as the number card shows, then press blue to confirm.

What the scoring code actually checks

```
repeat(press(red_left), n0)
press(blue)                                  Not in instr.
repeat(press(red_middle), n1)
press(blue)   # exactly twice                Not in instr.
both_arms_home()                             Not in instr.
```

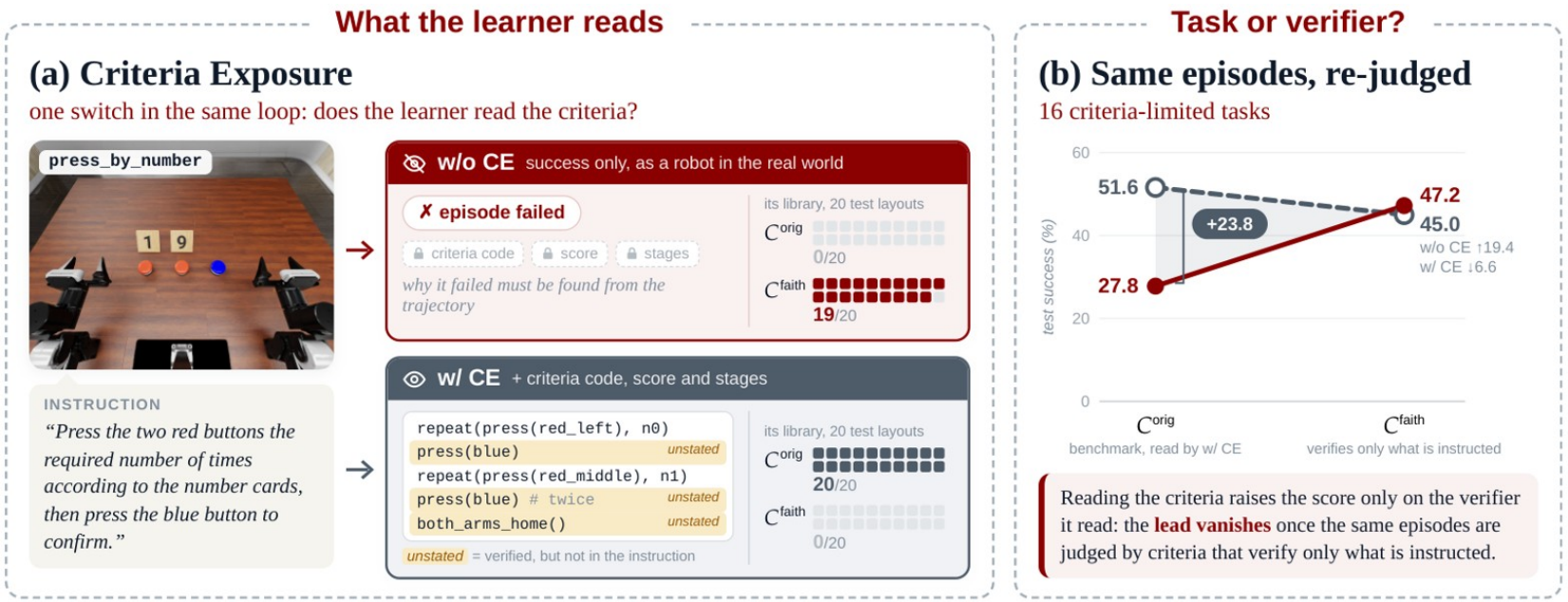
Criteria exposed to the learner by existing self-improvement methods (least→most), termed Criteria Exposure (CE)

Rule descriptions → Public rubrics → Numeric thresholds → Formal goal predicates → Verifier source code

Contrast: real-world robots get only instructions, at most plus success judgments from humans or a VLM. Imitation-learning demos already encode these details, so they are harmless there; an instruction-only coding agent cannot know them.

Research Question

Does reading criteria teach the agent the task or the verifier?



How to read Fig. 1

(a) Same loop; only whether the learner reads criteria differs

- w/o CE library: C_orig 0/20, C_faith 19/20
- w/ CE library: C_orig 20/20, C_faith 0/20

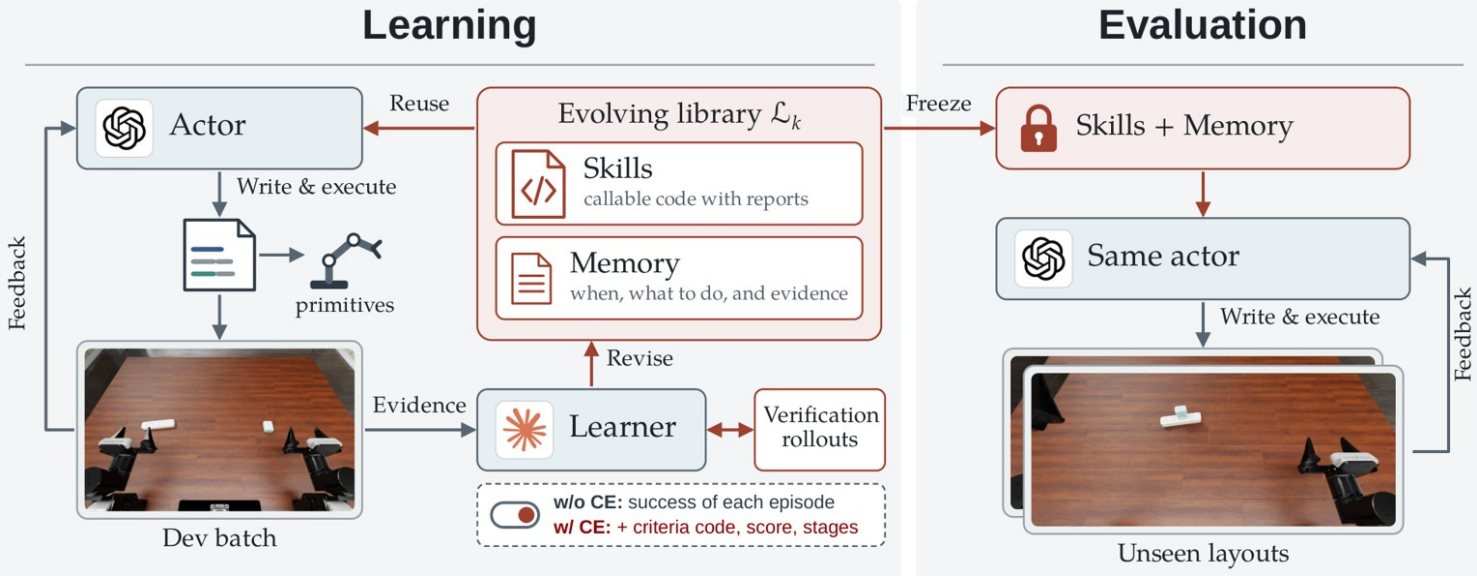
(b) 16 tasks; the same episodes scored by two sets of criteria

- Under C_orig: 51.6 vs 27.8, a 23.8 lead
- Under C_faith: 45.0 vs 47.2, lead vanishes

C_orig: original benchmark criteria; C_faith: authors' rewrite verifying only what the instruction states

Method

Design: same self-improvement loop, flipping only one switch



Actor (frozen)	GPT-6 Astra writes code over PyRUA-Lean primitives; RGB-only perception
Learner	Claude Code (Opus 5.5); each round rewrites evidence into a new skill + memory library
Learning	Per task: 5 rounds × 5 dev layouts (seed 1); roll back on clear regression
Testing	Library frozen, read-only; 20 unseen layouts (seed 0)

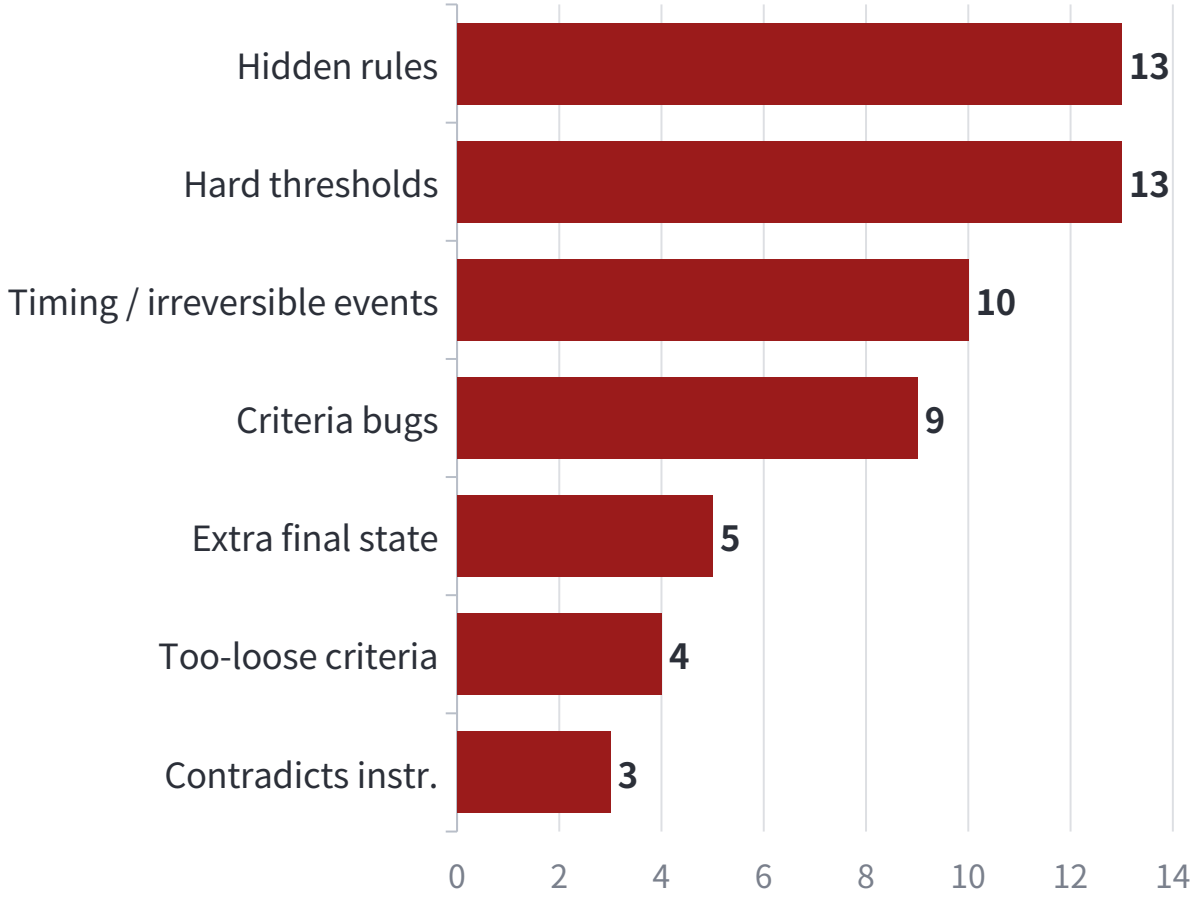
w/o CE Episode success only (+ trajectories)
Closest to the real-world setting

w/ CE Also reads criteria code, score, stages
The actor never sees these

Figure: paper Fig. 2; setup: paper § 3.1, Table 3

Criteria audit: 20 of 42 RoboDojo tasks mismatch instructions

Tasks per mismatch type (medium / high severity)



20 / 42 tasks have criteria inconsistent with instructions (26 variants total), each manually verified

33 vs 5 33 tasks' criteria require both arms to return home, but only 5 instructions mention it

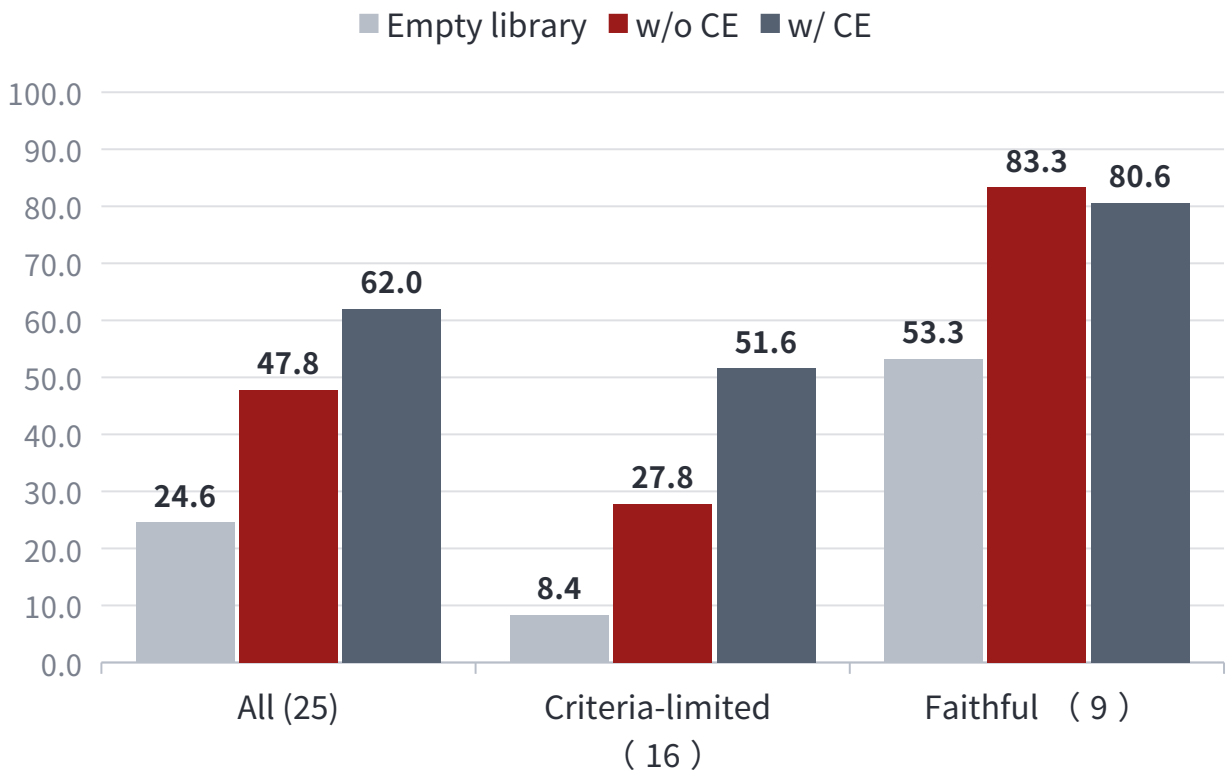
16 + 9 Experiments use 25 tasks: 16 criteria-limited + 9 faithful, grouped in advance

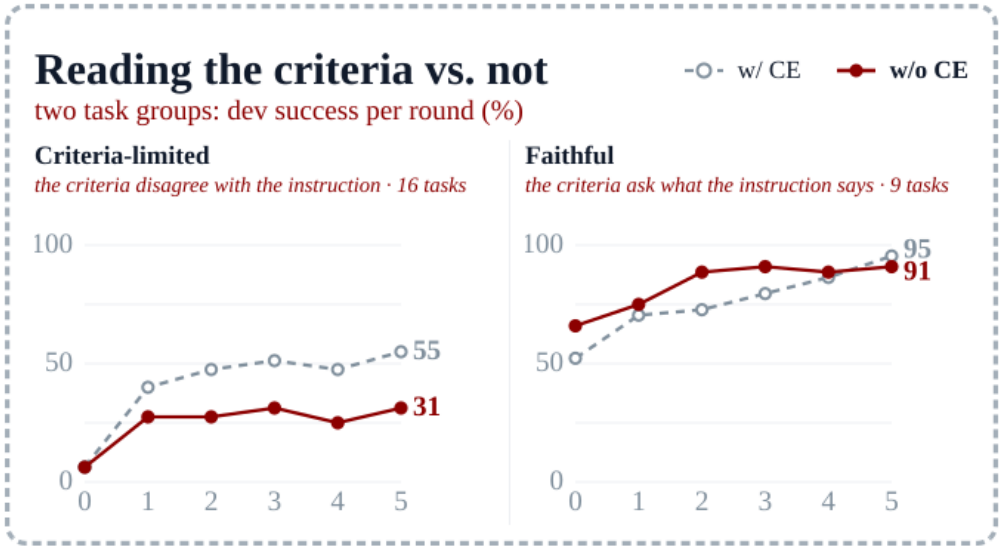
How instruction-faithful C_faith is written

- Edit scoring code only, not instructions; keep both-arm homing
- Remove/relax hidden rules, fix bugs, relax hard thresholds
- Tighten overly loose criteria where sim state reliably distinguishes
- Every verdict-flipped episode checked by hand; no misjudgments found

Data: paper § 3.2–3.3, Appendix B Table 5, Appendix C

H1: CE gains appear only on tasks with mismatched criteria





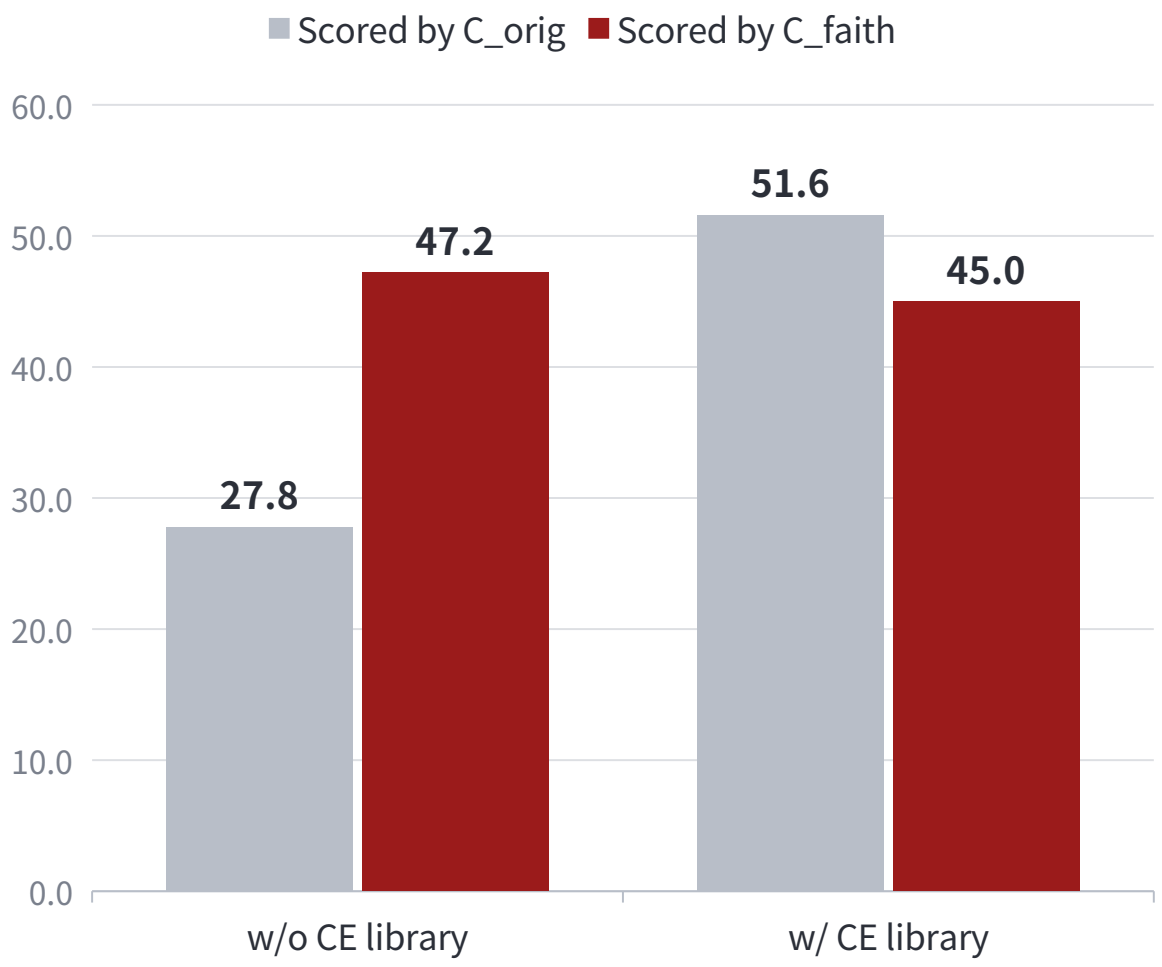
Dev success over rounds: the criteria-limited group splits from round 1 (40 vs 28), ending 55 vs 31; both exceed 90% on the faithful group.

Success-only feedback already doubles success (24.6→47.8); CE's extra **14.2 points** all come from criteria-limited tasks; on faithful tasks the two tie.

- **build_tower:** leftover blocks reveal the rule; both 20/20
- **press_by_number:** the button protocol is invisible in failed trajectories; the success-only learner never found it

Data: paper Table 1; figure: paper Fig. 3

H2: Re-scoring the same episodes faithfully erases CE's lead



+23.8 → −2.2

CE's lead shrinks by 25.9 points (per-task bootstrap, 95% CI 4.7–54.4)

93 : 17 → 47 : 54

Paired layouts: "CE-only" vs. "w/o CE-only" success counts fall within noise after the swap

87 / 89

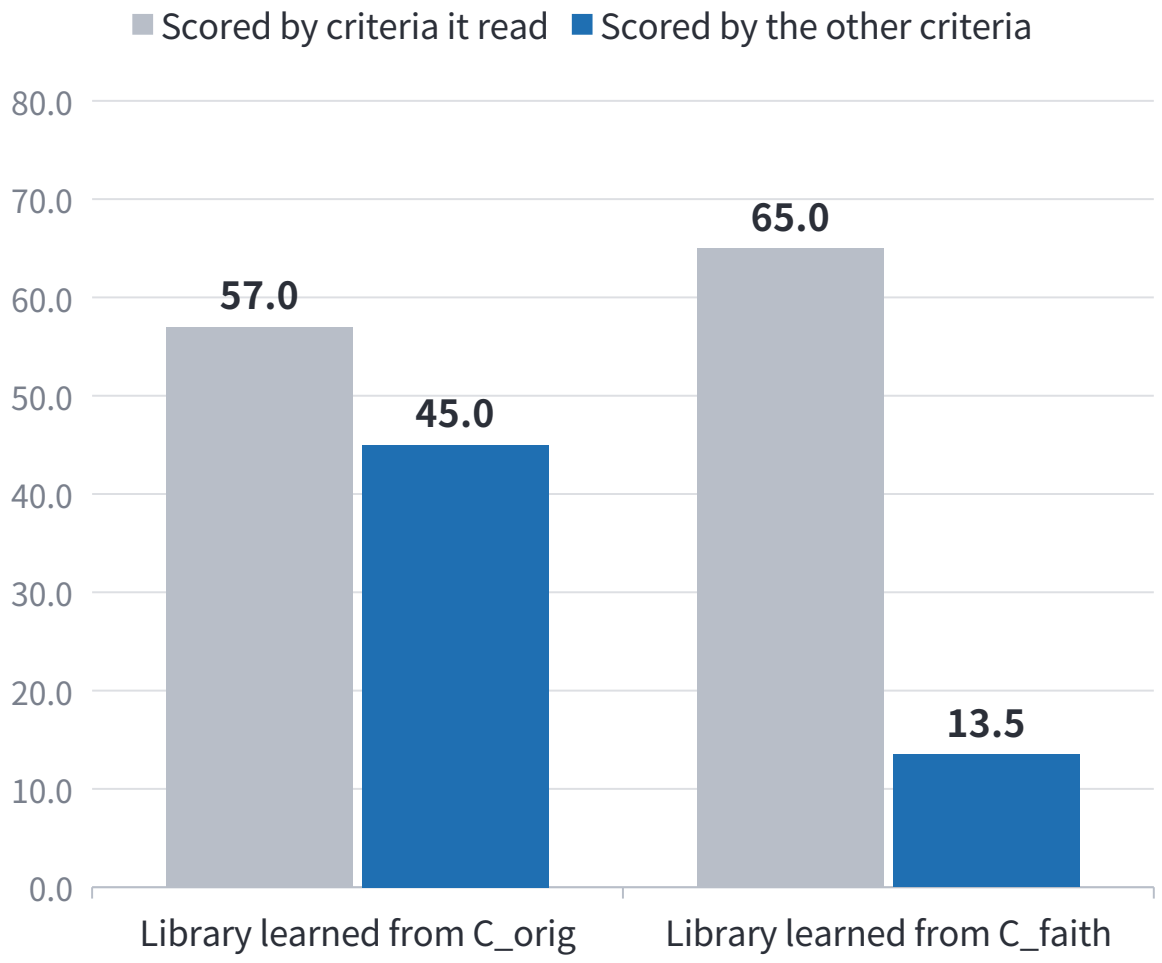
Of w/o CE's 89 successes under C_orig, C_faith rejects only 2

20/20 → 0/20

CE library on press_by_number; the w/o CE library goes from 0/20 to 19/20

Data: paper § 4.2.1, Fig. 1b / Fig. 5 (left)

H2: Each CE library scores high only under criteria it read



−12.0 vs −51.5: why asymmetric?

C_faith mainly removes or relaxes rules. Moving a C_orig library to C_faith loosens the standard; moving a C_faith library back to C_orig confronts it with a whole set of unseen rules.

Scores follow the criteria read, even looser ones

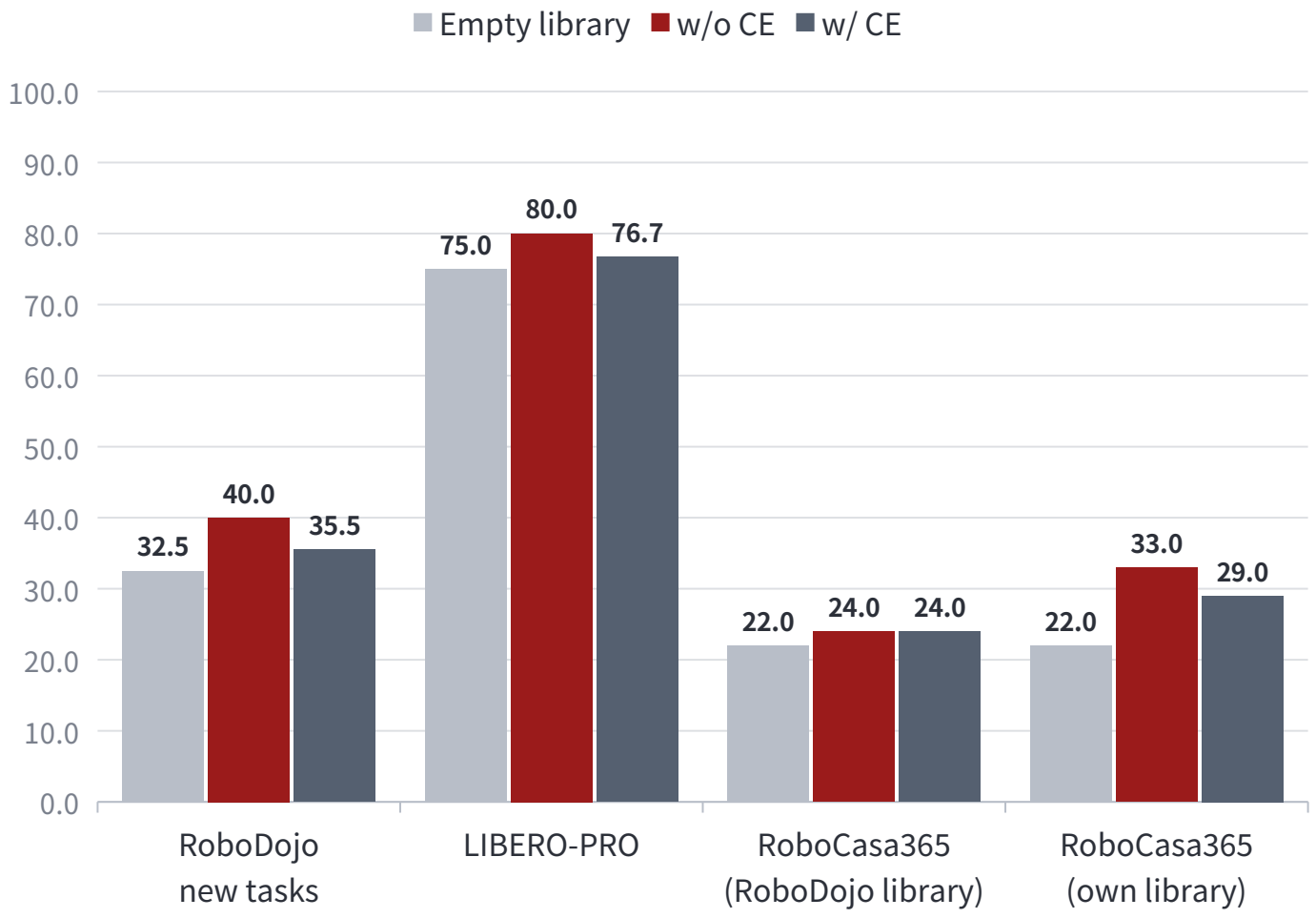
pour_by_language: the C_faith-reading library no longer uprights the bottle (required by C_orig): 19/20 under C_faith, 0/20 under C_orig.

CE also changes the lessons written into the library

About **1/3** of CE memory entries cite criteria code directly, vs. none for w/o CE; success-only learners must find physical causes.

Data: paper § 4.2.2–4.2.3, Fig. 5 (right)

H3: CE's edge doesn't transfer; only in-benchmark experience helps



Target (library source)	Empty library	w/o CE	w/ CE
RoboDojo （RoboDojo）	65/200	80	71
LIBERO-PRO （RoboDojo）	45/60	48	46
RoboCasa365 （RoboDojo）	22/100	24	24
RoboCasa365 （RoboCasa365）	22/100	33	29

- **Transfer to 10 new RoboDojo tasks:** w/o CE does better (+15 vs +6); CE's 14.2-point lead on source tasks does not reappear
- **Cross-benchmark:** RoboDojo libraries yield almost no gain on two new benchmarks
- **Explanation:** CE lessons encode "conditions and precision needed to pass a given criterion," which become noise in new settings; only physical regularities (e.g., gripper behavior) generalize

Takeaway: hide verifiers from learners in self-improvement evals

01

CE gains = benchmark flaws

CE raises scores only where instructions and criteria diverge, and its gain grows with that gap; it measures benchmark flaws, not agent capability.

02

No criteria exposure by default

Give learners only feedback a real robot can get (success/failure), and also report success under instruction-faithful criteria; only gains surviving the swap can be expected at deployment.

03

Spell out checks in instructions

Benchmarks should state counts, order, final states and arm homing in instructions, and publish versioned criteria plus a criteria–instruction mapping table.

The flip side: success-only self-improvement is real: success doubles and holds after swapping criteria.

Assessment

Verdict: clean design; C_faith and sample size need validation

Strengths

- Targets eval norms: turns "does the learner read criteria" from an implementation detail into a controlled variable
- Single-switch controlled study: actor, primitives, rounds and layouts all fixed
- Same trajectory scored by two criteria, cleanly separating behavior from criteria differences
- Counterfactuals in both directions (scoring vs. read criteria) corroborate each other
- The accompanying 42-task criteria audit is itself a useful community contribution

Limitations

- C_faith is author-written and mostly relaxes; some calls are subjective (e.g., "what counts as a tower") and need independent review
- Only 20 test layouts per task; wide aggregate CI (4.7–54.4); distillation variance not estimated; no ablations for mechanism claims
- Only two extremes compared (no exposure vs. source code); intermediate levels not measured
- Single actor/learner pair; criteria rewritten only for RoboDojo; sim success signals are cleaner than real-world ones
- Some rules are public on the official site; whether they count as legitimate task knowledge is debatable

“A robot in the real world never sees its verifier; the agent we measure should not either.”

Real robots never see their verifiers; agents we evaluate shouldn't either.

— Paper § 6 Conclusion

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